Name:Nidhi Patil
Cybersecurity Intern
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Task 11

NETWORKING FUNDAMENTALS

Networking Fundamentals Report

1. Introduction

Computer networking is the process of connecting two or more devices to communicate and share information. Networking is an important part of cybersecurity because it helps us understand how data travels between devices and how networks can be protected from cyberattacks.

2. Objectives

- Understand IP addresses and MAC addresses.
- Learn about network ports and protocols.
- Identify common network devices.
- Understand the role of routers, switches, and firewalls.
- Develop basic networking knowledge required for cybersecurity.

3. IP Address

An IP (Internet Protocol) address is a logical address assigned to a device on a network. It helps identify the device and allows data to reach the correct destination.

Types:

- IPv4: Uses 32 bits, for example, `192.168.1.10`.

- IPv6: Uses 128 bits, for example, `2001:db8::1`.

IP addresses can also be classified as public or private and static or dynamic.

4. MAC Address

A MAC (Media Access Control) address is a hardware-level address associated with a network interface. It is used for communication within a local network.

Example: `00:1A:2B:3C:4D:5E`

An IP address is used for logical addressing, while a MAC address is used for link-layer communication on a local network.

5. Network Ports

A port is a logical number used by transport protocols such as TCP and UDP to direct network traffic to the appropriate application or service.

Common ports:

- Port 20/21 – FTP data/control
- Port 22 – SSH
- Port 25 – SMTP
- Port 53 – DNS
- Port 80 – HTTP
- Port 443 – HTTPS
- Port 3389 – Remote Desktop Protocol (RDP)

Open ports should be monitored and protected because unnecessary exposed services can increase security risks.

6. Network Protocols

A network protocol is a set of rules that allows devices to communicate.

- **TCP:** Provides reliable, ordered delivery of data.
- **UDP:** Sends data without establishing a TCP-style connection.
- **HTTP/HTTPS:** Used to access websites; HTTPS protects data using TLS.
- **DNS:** Translates domain names into IP addresses.
- **DHCP:** Automatically assigns IP configuration to devices.
- **SSH:** Enables secure remote access to systems.
- **ARP:** Helps discover the link-layer address associated with an IPv4 address on a local network.

7. Network Devices

Router

A router forwards packets between different networks. Home routers commonly connect a local network to the internet.

Switch

A switch connects devices within a local area network and forwards Ethernet frames using MAC addresses.

Firewall

A firewall filters network traffic according to security rules. It helps block unauthorized connections and reduce network risks.

Modem

A modem connects a network to an internet service provider through the provider's access technology.

8. Basic Networking Example

When a user opens a secure website:

1. DNS helps find the website's IP address.
2. The device sends network traffic toward the destination through the router.
3. HTTPS commonly uses TCP port 443, with TLS protecting the communication.
4. Network devices forward the traffic toward the website.
5. Firewalls and other security controls can filter traffic according to their rules.

9. Importance of Networking in Cybersecurity

Networking knowledge helps cybersecurity professionals:

- Identify suspicious network traffic.
- Understand how attackers may exploit exposed services.
- Configure firewalls and network access controls.
- Investigate network-related incidents.
- Monitor ports, protocols, and connected devices.
- Improve network security and troubleshooting.

10. Tools Used

- **Web browser:** For studying networking concepts and protocols.
- **Cisco Packet Tracer:** For creating and simulating network topologies and testing basic connectivity.

11. Security Best Practices

- Use strong passwords and multi-factor authentication.
- Keep routers, firewalls, and network devices updated.
- Disable unnecessary services and close unused ports.
- Use HTTPS and secure remote-access protocols.

- Monitor network traffic and review firewall rules.
- Use authorized lab environments for network testing.

12. Conclusion

This task introduced the fundamental concepts of computer networking, including IP addresses, MAC addresses, ports, protocols, and network devices. These concepts form the foundation of cybersecurity and help in understanding network communication, identifying risks, and protecting systems against unauthorized access.

Interview Questions

1. What is an IP address?

An IP (Internet Protocol) address is a logical address assigned to a device connected to a network. It helps identify the device and allows data to reach the correct destination.

Example: `192.168.1.10`

2. What is a MAC address?

A MAC (Media Access Control) address is a hardware-level address associated with a device's network interface. It is used for communication within a local network.

Example: `00:1A:2B:3C:4D:5E`

3. What is a port?

A port is a logical number used to direct network traffic to the correct application or service on a device.

Examples:

- Port 80 — HTTP

- Port 443 — HTTPS
- Port 22 — SSH